

When Routing Entropy Tracks Length, Not Complexity: A Cross-Model Token-Position Confound in MoE Interpretability

Jeffrey Shorthill*

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Abstract

Mixture-of-experts (MoE) language models send each token to a small subset of experts. The entropy of that routing distribution (how evenly a token’s weight spreads across experts) is a convenient one-number summary, and a higher value invites reading as a sign that the model is working harder. Across a 168-prompt suite graded into twelve levels of intended cognitive complexity, the routing entropy averaged over all prefill tokens rises with the intended level in two MoE models (DeepSeek V3.1, Spearman $\rho = +0.80$; Qwen3.5-397B, $\rho = +0.62$; $n = 168$ each). The same averaged quantity tracks prompt token count more strongly ($\rho = +0.88$ and $+0.78$). Routing entropy rises with token position inside the prefill, so longer prompts, which the complexity grading also makes longer, collect more high-entropy late positions and lift the all-token mean. Reading routing entropy at the final prefill token, whose receptive field already spans the whole prompt, removes the gradient: the correlation with level falls to $\rho = +0.02$ ($p = 0.82$) for DeepSeek V3.1 and $\rho = -0.06$ ($p = 0.42$) for Qwen, and a level-1 versus level-12 comparison reverses sign, with the short prompts showing slightly higher final-token entropy. The reversal holds at $p < 10^{-3}$ in both models, and the all-token pattern repeats in a DeepSeek R1 run on the same suite. In this setup the aggregate gradient reflects token count and token position. We describe the readout that removes it, the mechanism behind the inflation, and the length-matched follow-up that would settle what remains.

Keywords: mixture-of-experts; routing entropy; token position; prompt-length confound; prefill; interpretability.

1 Introduction

A routing analysis of a mixture-of-experts model starts from a simple object: for each token, the router produces a distribution over experts and keeps the top few. Summary statistics of that distribution are easy to compute and easy to compare across prompts. Routing entropy—the spread of the per-token routing distribution—is one such statistic. It is appealing as a workload readout because a peaked distribution looks like confident specialization and a flat distribution looks like the model hedging across many experts.

*Independent researcher. Correspondence: jws299792@icloud.com. This is version 1.0 (June 2026), released under CC BY 4.0. Data, code, and a claim-by-claim source index accompany the paper; see the Data and Code Availability statement.

This note follows one such reading and reports where it breaks. We graded a prompt suite into twelve levels meant to span increasing cognitive demand, from rote repetition up to layered self-referential prompts, and measured routing entropy on two MoE models. Averaged over the tokens of each prompt, routing entropy climbs with the intended level. Taken alone, that gradient supports treating the statistic as a complexity proxy.

The gradient has a second explanation. The graded prompts also grow longer with level, and the averaged statistic tracks prompt length at least as tightly as it tracks the grading. The reason is positional: within a single prefill pass, later token positions carry higher routing entropy on average, because under causal attention they read a larger context. Averaging entropy over all positions gives long prompts more weight from their high-entropy tail, so length raises the mean on its own. Moving the measurement to the final prefill token, a single position whose receptive field already covers the whole prompt, removes the positional averaging. Under that readout the gradient with level disappears in both models, and the extreme levels swap order.

The result is narrow. It concerns one prompt suite, one pair of models, and one way of aggregating a routing statistic. The point to carry away is mechanical: when routing entropy is averaged over prompts of different lengths, prompt length can stand in for whatever the prompts were meant to vary.

The contribution is a measurement caution with a worked diagnosis. We document the apparent complexity gradient in the all-token mean across two MoE models and a replication; we show the same statistic tracks prompt token count at least as tightly; we trace the inflation to a positional effect, that routing entropy rises with token position within the prefill; and we give a position-controlled readout, the final prefill token, under which the gradient with intended level disappears and the extreme levels swap order.

2 Related work

Mixture-of-experts routing and its summary statistics. A mixture-of-experts layer routes each token through a learned gate to a few of many expert subnetworks (Jacobs et al., 1991; Shazeer et al., 2017), a design scaled into large transformers by Switch Transformers (Fedus et al., 2022) and, in the models studied here, by the fine-grained-expert routing of DeepSeek (Dai et al., 2024; DeepSeek-AI, 2024, 2025) and the Qwen3 lineage (Yang et al., 2025). Because the router emits an inspectable distribution at every token, summary statistics of that distribution are a common readout; routing analyses of this kind go back at least to the per-token study of Mixtral (Jiang et al., 2024), and the Shannon entropy of the routing weights is one such one-number summary.

Routing signals read as difficulty or effort. A line of work treats the router’s distribution as a difficulty signal. Huang et al. (2024) make this explicit, activating more experts for inputs the router is less confident about, on the premise that harder inputs warrant more computation; mixture-of-depths (Raposo et al., 2024) routes a per-token compute budget on a related premise. Reading a flat (high-entropy) routing distribution as a sign that the model is working harder is the intuition this note examines. We do not dispute that the router responds to input; we show that a between-prompt gradient in *averaged* routing entropy can be produced by prompt length and token position, so an aggregate gradient is weak evidence on its own that the experts are responding to intended complexity.

Position-dependent behavior and length confounds. Causal self-attention (Vaswani et al., 2017) makes a token’s representation depend on every earlier position, so later positions are computed over more context, and transformers are known to behave sharply differently across positions, for instance the attention sinks at initial positions documented by Xiao et al. (2024). The mechanism we report is in this family: routing entropy rises with prefill position, so averaging over positions over-weights the high-entropy tails of long prompts. That length can stand in for an intended factor is a familiar hazard elsewhere; in reinforcement learning from human feedback, for example, a length-only proxy reproduces much of an apparent quality gradient (Singhal et al., 2024). The same hazard appears here for a routing statistic. A companion analysis by the present author separates the prefill and generation passes for a different routing question, expert–domain specialization (Shorthill, 2026); the prefill capture and prefill/generation framing are shared, the measured quantity is not.

3 Why routing entropy is tempting as a complexity readout

Three properties make routing entropy attractive as a between-prompt measure.

It is cheap and intrinsic. The router already computes the distribution during a forward pass, so capturing it adds no probes, no fitted readout, and no labels. A prefill-only pass over a prompt set yields a number per token at every MoE layer.

It has a clean interpretation per token. Low entropy means the token’s weight concentrates on a few experts; high entropy means it spreads out. Read locally, that is a statement about how committed the router is at that position, and it is tempting to extend the local reading to a whole prompt by averaging.

It moves with things researchers care about. In the suite studied here, the all-token mean rises smoothly across the twelve graded levels in both models (Figure 1, circles). A monotonic climb across a deliberately ordered factor is the kind of pattern that invites a one-line summary: “harder prompts route more diffusely.”

The averaging step is where the reading becomes fragile. A per-token quantity becomes a per-prompt quantity only after a choice of how to pool positions, and the mean over all prefill tokens is not neutral with respect to prompt length. The rest of this note works through that step.

4 Methods

Models. We use two mixture-of-experts models. DeepSeek V3.1 (DeepSeek-AI, 2024) has 256 experts with 8 active per token across 58 MoE layers; Qwen3.5-397B-A17B (Yang et al., 2025) has 512 experts with 10 active across 60 MoE layers. Both were run from quantized weights (UD-Q2_K_XL) on a llama.cpp fork (build b8123, `capture_activations`), with greedy decoding and routing captured in prefill-only mode (`n_predict` = 0). A DeepSeek R1 run (DeepSeek-AI, 2025) on the same suite serves as a replication. The router output recorded at each prefill token, for each MoE layer, is the input to every measurement below.

Prompt suite. The suite holds 168 prompts: 14 prompts at each of 12 levels. The levels are an intended-complexity ordering, from rote repetition and factual recall at the low end, through logical reasoning, analogy, theory-of-mind, and ethical dilemmas, up to self-referential and persona-style

prompts at the high end. The levels were written to vary in cognitive demand. They also vary in length: the low levels are short (mean near 30 tokens) and several high levels run long (the longest level averages about 150 tokens). That length structure is the confound this note tracks.

Routing entropy. At a token, each MoE layer’s router yields weights over its experts; we take the Shannon entropy of that weight distribution, average across the model’s MoE layers, and normalize to the unit interval, giving a per-token routing entropy in $[0, 1]$. We summarize a prompt two ways. The *all-token mean* averages the per-token value over every prefill position. The *final-token* readout takes the value at the last prefill position only. Under causal attention the final position has attended to the entire prompt, so its routing entropy is a single-position summary that does not average over the length of the prompt.

Statistics. For each model we compute Spearman rank correlations over all $n = 168$ prompts between routing entropy and (i) the intended level and (ii) the prompt token count, for both the all-token and final-token readouts. We compare the lowest and highest levels (L1, $n = 14$; L12, $n = 14$) with a Wilcoxon rank-sum test. The intended-level correlation is the quantity a complexity reading would lean on; the token-count correlation is the confound; the final-token readout is the position-controlled check.

5 Results

5.1 The all-token mean rises with the intended level

Averaged over all prefill tokens, routing entropy increases across the twelve levels in both models (Figure 1, circles). The Spearman correlation with level is $\rho = +0.80$ ($p \approx 5.6 \times 10^{-39}$) for DeepSeek V3.1 and $\rho = +0.62$ ($p \approx 5.7 \times 10^{-19}$) for Qwen (Table 1). Read on its own, this is the gradient a complexity proxy would predict.

5.2 The same average tracks token count more strongly

The graded levels grow longer toward the high end, and the all-token mean follows length closely. Across all 168 prompts it correlates with token count at $\rho = +0.88$ ($p \approx 1.8 \times 10^{-55}$) for DeepSeek V3.1 and $\rho = +0.78$ ($p \approx 8.3 \times 10^{-36}$) for Qwen. In both models the correlation with token count is tighter than the one with level. Figure 2 shows the per-level means against per-level mean token count for DeepSeek V3.1: the entropy ordering and the length ordering largely coincide. Token count and intended level are themselves strongly related in this suite, so a measure that rises with length will also rise with level.

5.3 The final-token readout removes the gradient

Measured at the final prefill token, routing entropy no longer orders by level (Figure 1, squares). The correlation with level falls to $\rho = +0.02$ ($p = 0.82$) for DeepSeek V3.1 and $\rho = -0.06$ ($p = 0.42$) for Qwen (Table 1). The final-token correlation with token count is also small ($\rho = +0.16$ and $\rho = -0.22$), in line with a single-position readout that does not pool over length. Figure 3 places the four correlations side by side: large and positive for the all-token readout, near zero for the final-token readout, in both models.

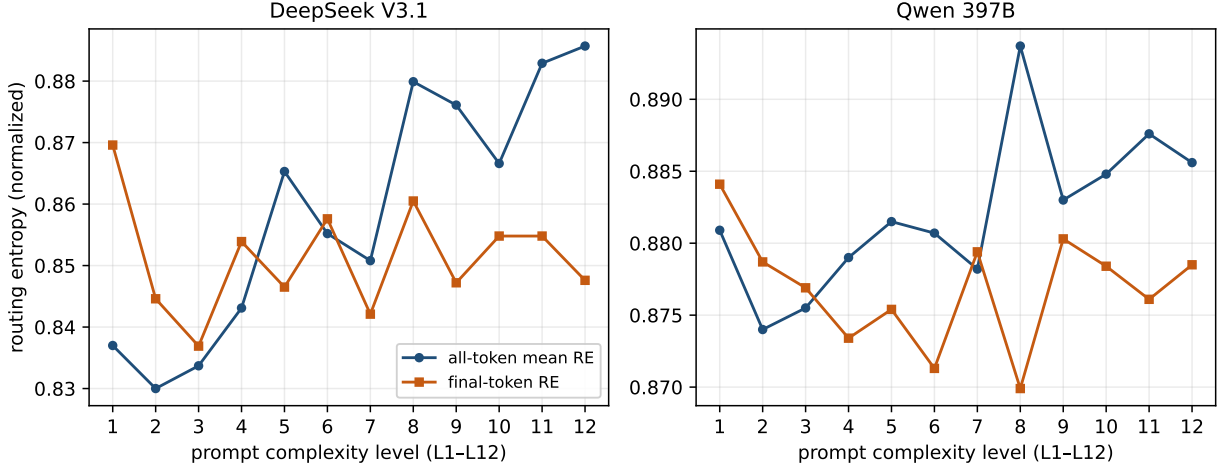


Figure 1: Per-level mean routing entropy against intended complexity level, for DeepSeek V3.1 (left) and Qwen 397B (right). Circles: all-token mean routing entropy. Squares: final-token routing entropy. Each point averages 14 prompts; 168 prompts per model. The all-token mean climbs with level in both models; the final-token readout stays within a narrow band with no monotonic trend.

Table 1: Spearman correlations of routing entropy with intended level and with prompt token count, by readout and model ($n = 168$). The all-token mean orders by level and by length; the final-token readout does neither.

Model	Readout	vs. level (ρ , p)	vs. token count (ρ , p)
DeepSeek V3.1	all-token	+0.80 (5.6×10^{-39})	+0.88 (1.8×10^{-55})
DeepSeek V3.1	final-token	+0.02 (0.82)	+0.16 (0.037)
Qwen 397B	all-token	+0.62 (5.7×10^{-19})	+0.78 (8.3×10^{-36})
Qwen 397B	final-token	-0.06 (0.42)	-0.22 (0.0042)

5.4 The extreme levels swap order

A direct comparison of the lowest and highest levels shows the same thing as a direction change. Under the all-token mean, level 12 sits above level 1 in both models (DeepSeek V3.1, $W = -4.00$, $p = 6.4 \times 10^{-5}$; Qwen, $W = -2.94$, $p = 3.3 \times 10^{-3}$). Under the final-token readout the order flips: level 1 sits above level 12 (DeepSeek V3.1, $W = +3.77$, $p = 1.7 \times 10^{-4}$; Qwen, $W = +3.31$, $p = 9.4 \times 10^{-4}$), shown in Table 2. At a single position that already covers the whole prompt, the short, low-level prompts carry slightly higher routing entropy than the long, high-level ones. The gap is larger in DeepSeek V3.1 (about 0.022) than in Qwen (about 0.006).

5.5 The pattern is consistent across runs

A DeepSeek R1 run on the same 168-prompt suite reproduces the all-token picture: routing entropy correlates with level at $\rho = +0.84$ and with token count at $\rho = +0.86$, again ordering by length at least as well as by level. The earlier 98-prompt baseline on DeepSeek V3.1 showed the same signature on the generation side, where the all-token mean correlated with generated length ($\rho = +0.51$, $p \approx 5 \times 10^{-7}$) more clearly than with level ($\rho = +0.20$, $p = 0.05$). The positional dependence behind these patterns was checked directly with a per-token diagnostic, in which routing entropy rises with

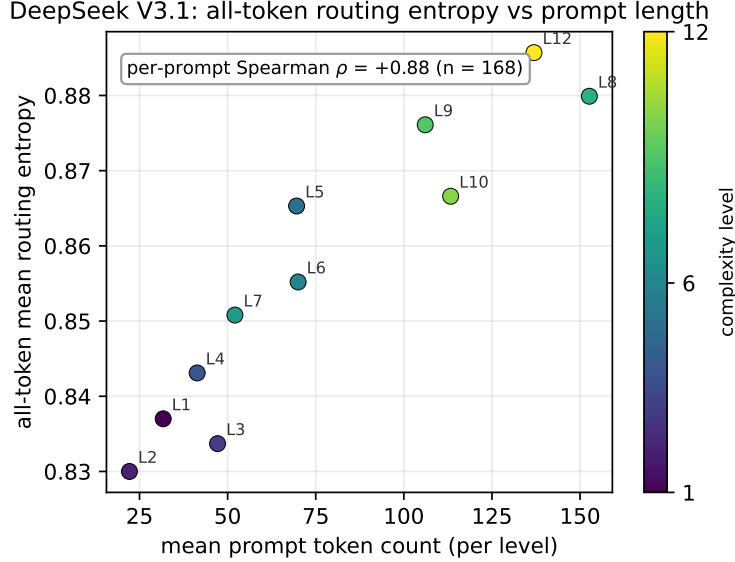


Figure 2: DeepSeek V3.1 per-level all-token mean routing entropy against per-level mean prompt token count; markers labeled and colored by level. The annotated correlation is the per-prompt Spearman value over all 168 prompts. All-token mean routing entropy and prompt length rise together.

Table 2: Level-1 versus level-12 routing entropy (Wilcoxon rank-sum, $n = 14$ per level). The all-token mean places the complex level higher; the final-token readout places the simple level higher.

Model	Readout	L1 mean	L12 mean	Direction (p)
DeepSeek V3.1	all-token	0.837	0.886	L12 > L1 (6.4×10^{-5})
DeepSeek V3.1	final-token	0.870	0.848	L1 > L12 (1.7×10^{-4})
Qwen 397B	all-token	0.881	0.886	L12 > L1 (3.3×10^{-3})
Qwen 397B	final-token	0.884	0.879	L1 > L12 (9.4×10^{-4})

token position within the prefill.

A separate, tightly templated prompt suite gives a preliminary cross-check in the other direction. From a forced-choice triage set with uniform scaffolding and a narrow length band (227–245 tokens), only 17 of 32 prompts survived as raw captures, all in the three lowest levels. On that subset the all-token mean correlates negatively with both token count ($\rho = -0.66$) and level ($\rho = -0.58$), and token count again accounts for more of the variance than level. Because the subset is small and spans only three levels, we treat it as a preliminary note that the readout stays sensitive to prompt construction, not as a second gradient.

6 Interpretation

Why averaging over positions favors long prompts. Under causal attention (Vaswani et al., 2017), the representation at a prefill position depends on every earlier position, so later tokens are computed over more context. In both models the router responds to that by spreading weight a little more widely at later positions, which raises per-token routing entropy with position. The all-token mean weights every position equally, so it collects more high-entropy positions from a long

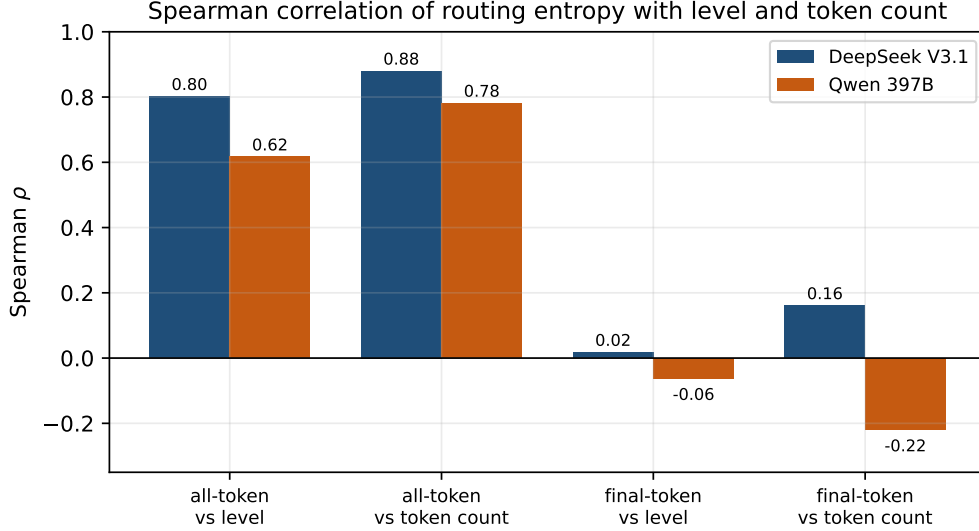


Figure 3: Spearman correlation of routing entropy with intended level and with prompt token count, for each readout and model ($n = 168$). The two left groups use the all-token mean; the two right groups use the final-token readout. The correlation with level is large under all-token aggregation and near zero at the final token.

prompt than from a short one. When the prompt set is ordered so that the intended-complexity levels also grow longer, the mean rises with level on length alone. The final-token readout avoids the pooling step: it reports one position whose context is the full prompt, so every prompt contributes a single value regardless of length.

The reversal. At the final token, the short low-level prompts carry slightly higher routing entropy than the long high-level prompts, and the difference is consistent across both models. A plain reading is that with less prior context to condition on, the router at the final position spreads weight a little more widely. We offer that as a description of the measured direction; the mechanism stays open. The effect is small (about 0.02 and 0.006 in normalized entropy), and a stronger account would need a position-resolved analysis.

Scope of the claim. What these runs support is specific. In this suite and these two models, the all-token mean routing entropy orders prompts by length and by token position, and a position-controlled readout does not reproduce the ordering by intended complexity. The finding speaks to one way of turning a per-token routing statistic into a per-prompt number when prompts differ in length. It does not establish that routing entropy carries no information about prompt content, and it does not bear on routing analyses that hold length fixed or that model position explicitly.

7 Limitations and falsification tests

Limitations. The intended-complexity grading and prompt length are entangled in this suite, so these runs separate position from the all-token average but do not yet separate length from complexity; a length-matched suite is the missing control. The final-token readout removes the

positional averaging at the cost of resting on a single position, which carries its own sampling noise. The proposed reason for the final-token reversal is a description of direction, not a demonstrated mechanism. The evidence covers two model families plus an R1 replication, all from quantized weights under one inference setup; other architectures and precisions are untested here. The result diagnoses one measurement setup; an all-token average stays useful when position and length are modeled directly.

Tests that would weaken or overturn this reading. The following are proposed, not yet run.

- **Length-matched complexity prompts.** Hold token count fixed across levels. If the all-token gradient with level survives at matched length, the token-count explanation weakens.
- **Fixed-token-budget variants.** Pad or trim prompts to a common budget and re-measure both readouts.
- **Within-position entropy curves by level.** Compare entropy-versus-position curves across levels; overlapping curves would attribute the all-token gap to position, separated curves would not.
- **Final-token plus mean-of-last- k .** Compare the single final position with an average over the last k positions to gauge single-position noise against the positional control.
- **More MoE families.** Repeat on additional architectures and precisions to test whether the positional dependence is general.
- **Token-count-preserving paraphrase.** Shuffle or paraphrase prompts within a level while holding token count, to separate wording from length.
- **Mixed-effects regression.** Model routing entropy with token count, position, model, and level together, and ask whether level retains an effect once the first three are included.

8 Conclusion

In this setup, aggregate routing entropy tracks token count and token position. The twelve-level gradient that looked like a complexity signal under the all-token mean follows prompt length at least as closely, and it comes apart at the final prefill token, where the ordering by level disappears and the extreme levels change places. The same all-token signature appears across DeepSeek V3.1, a DeepSeek R1 replication, and Qwen 397B. The practical takeaway is concrete: when routing entropy is compared across prompts of different lengths, report it at a position-controlled readout, or model position and length directly, before reading a between-prompt difference as a difference in what the prompts demand. A length-matched suite would settle how much, if any, of the level effect remains once length is held fixed.

Reproducibility and audit note

The measurements above are recomputed in this paper’s `make_figures.py` from files in a preserved archive (`token-confound-archive/`). The per-level means and summary correlations are transcribed from the archive’s evidence summary (`CROSS_MODEL_POSITION_CONFOUND.md`), which matches the project journal’s record of the same run. Per-level token counts are computed from

the DeepSeek V3.1 per-prompt summary (`data/ds31-168q-1_prefill.json`); the DeepSeek R1 per-prompt routing entropy and token counts are read from `raw/168q-r1-deepseek-r1/results_168_r1_prefill.json`. The per-token position diagnostic is in `data/diagnostic_results.json`. A source-to-claim index is given in `SOURCES.md`.

The archive keeps the trail that led here. An earlier generation-phase measurement on a 30-turn run reported a sharp peak on the final turn; that turn was also the longest generation, and the peak was set aside once length was recognized as its driver—the same length sensitivity that this note isolates on the prefill side. The archive also records a data-integrity event from that period: while assembling result tables, an automated step produced per-prompt rows that matched the correct aggregates but were not extracted from the run logs, and some per-prompt slope signs were wrong as a result. The tables were regenerated from the logs and a verify-against-log rule was added. The aggregate correlations reported in this paper are the verified, log-derived values, and the earlier per-prompt write-ups that predate the position control are retained in the archive as superseded.

Data and Code Availability

The measurements in this paper are recomputed from a preserved capture archive (`token-confound-archive/`) by `make_figures.py`, which fits the three figures and the reported correlations from the per-prompt routing-entropy and token-count records and re-runs no model. The per-level summaries and correlations are transcribed from the archive’s evidence summary (`CROSS_MODEL_POSITION_CONFOUND.md`); per-prompt token counts and the DeepSeek R1 replication values are recomputed from the archived per-prompt JSON, and the per-token position diagnostic from `data/diagnostic_results.json`. A claim-by-claim source index is given in `SOURCES.md`. No new model training was performed.

AI Collaboration Statement

Generative AI (Anthropic’s Claude) was used as a tool in preparing this manuscript: to assist with drafting and revising prose, organizing the manuscript, and locating and formatting the bibliography. All captures, statistics, and numerical results are the author’s; every reported value was traced to a source capture (see `SOURCES.md`), and every reference was verified by the author against its authoritative record to exist and to support the statement it is attached to. No AI system is an author. The author reviewed the entire text and takes full responsibility for its content, including any errors.

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This work received no external funding. The author declares no competing interests. The models studied are third-party releases used as-is for measurement; the author has no affiliation with their creators.

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